Response to Reviewers — Paper ID 1170

Dropout-Aware Training for Sensor-Robust Fault Localization
with Spatio-Temporal Graph Neural Networks

Dear Editor and Reviewers,

We thank you for the careful reading of our manuscript and for the constructive comments. We have revised the manuscript to address every comment. All changed or added text is marked in blue in the revised manuscript. Text that we removed cannot be marked in blue; it is listed in Part E. Page numbers refer to the revised manuscript.

Part A — Reviewer #1

We thank Reviewer #1 for the positive assessment, in particular that "the comparison across five architectures provides useful evidence about sensor-dropout effects," and for the helpful suggestions.

Comment 1.1

> *"Please clarify how the 25 μPMU locations and their topology-derived edges were selected."*

Response. We agree that the description was not precise enough. The 25 μPMU buses are taken unchanged from [1], and we now list them. Each unit measures the phases present at its bus in the feeder model, and the placement is not optimized in this work. We now state the exact edge rule used in our code: on a breadth-first spanning tree of the feeder rooted at the substation (bus 150), each μPMU is connected to the nearest μPMU upstream of it, except that a single- or two-phase unit whose nearest upstream unit is also single- or two-phase is not connected there. Two units (buses 33 and 113) are in that case; a fallback connects them to the μPMU at bus 1 so that the graph stays connected. We state that bus 1 is chosen by a fixed implementation rule (the first connected μPMU in bus-number order), not by electrical proximity. The graph itself is unchanged.

Change. Section 3.1, paragraph 2 ([p. X: s3.1-edges]); Fig. 1 caption ([p. X: fig1]).

Comment 1.2

> *"Consider evaluating additional feeders or topologies to test generalizability."*

Response. We agree that a single feeder limits generality. We did not add a new feeder in this revision, for two reasons. First, the study builds on the methodology of [1], and a direct comparison with it requires the same feeder and the same μPMU placement. Second, a new feeder requires a new simulated corpus and the retraining of five architectures, each with standard and dropout-aware training, at three seeds; this was not possible within the revision period. The proposed method changes only the input masking during training, so it can be applied to another feeder without modification; its effect on other feeders, however, is not measured, and we now state this clearly. The single-feeder limitation is the first item of the Limitations, and we name concrete next feeders.

Change. Section 5.3, first item ([p. X: s5.3-feeder]); Section 6, future work ([p. X: s6]).

Comment 1.3

> *"The conclusions appropriately acknowledge that the architecture rankings are uncertain with only three training seeds."*

Response. We agree, and we kept this caution. In the revision it is stated more precisely: every p-value is now Holm-adjusted within a stated family (Comment 3.2), the cross-architecture rankings are stated to be conditional on the spatial depth (Comment 2.5), and the spread between mask realizations within a seed is shown to be small compared with the spread between seeds (Comment 2.3).

Change. No separate change; see Comments 2.3, 2.5 and 3.2.

Part B — Reviewer #2

We thank Reviewer #2 for the detailed comments, which led to four new experiments.

Comment 2.1

> *"Please test a wider range of training dropout levels, including the evaluation maximum of 10 sensors."*

Response. We added this experiment. We retrained RGSAGE-Max with the training grid $\{0,1,2,5,10\}$ (DA-wide) at three seeds and re-evaluated it together with the original grid $\{0,1,2,5\}$ on identical masks. Adding $k = 10$ to the grid raises macro-F1 at $k = 10$ from 0.675 to 0.704 ($+0.030$) and lowers the clean macro-F1 from 0.839 to 0.833 (-0.006); AUDC rises from 0.913 to 0.928. Both differences have the same sign in all three seeds, but neither is significant after Holm correction ($p_{\text{Holm}} = 0.057$ for each). We therefore report a consistent trade-off at $n = 3$ and keep the original grid as the main method. DA-wide is now a row of Table 6.

Change. Section 3.5 ([p. X: s3.5]); Section 4.3, new paragraph "Training dropout grid" and footnote ([p. X: s4.3-grid]); Table 6, new row ([p. X: tab6]); Section 5.3 ([p. X: s5.3-grid]).

Comment 2.2

> *"The random sensor-loss protocol should be supplemented with correlated or geographically clustered failures."*

Response. We added a clustered missingness policy: the dropped μPMUs form a connected block of the measured-only graph, grown from a random root μPMU. We evaluated it on all five architectures, for standard and dropout-aware training, at three seeds. For standard RGSAGE-Max, clustered loss is milder than random loss (AUDC 0.553 against 0.402); per dropout level, only the difference at $k = 5$ is significant after Holm correction. For the dropout-aware model the two policies give nearly the same curve, and its gain over standard training under clustered loss is significant at every k ($p_{\text{Holm}} \leq 0.022$). Across the five architectures, the dropout-aware AUDC is higher under clustered loss for all five (descriptive), but the gain in absolute macro-F1 is significant at every k only for RGSAGE-Max. We report the milder effect of clustered loss as observed, not established: a connected block does not drop every μPMU equally often (per-node drop rates from 0.18 to 0.87 at $k = 10$, against 0.40 under random loss), so this comparison does not isolate spatial correlation. We state this in the Limitations and in future work.

Change. Section 3.4, policy definition ([p. X: s3.4-missing]); Section 4.3, new paragraph "Clustered dropout" ([p. X: s4.3-clustered]); Section 4.4, new paragraph ([p. X: s4.4-clustered]); Section 5.3 ([p. X: s5.3-clustered]); Section 6 ([p. X: s6]).

Comment 2.3

> *"Clarify whether statistical tests account for the three mask realizations as well as training-seed variation."*

Response. Yes. For each training seed, the three mask realizations are averaged first; the paired t-test then uses the three seed-level values (df $= 2$). Mask realizations are therefore not treated as independent samples. We now state this procedure explicitly, and we report that the spread between mask realizations within a seed (0.0002–0.0046 macro-F1) is small compared with the spread between seeds (0.008–0.061).

Change. Section 3.4, "Dropout sweep and metrics" ([p. X: s3.4-sweep]); Section 4.5, paragraph 1 ([p. X: s4.5-p1]).

Comment 2.4

> *"Compare the proposed augmentation with stronger learned imputation methods, not only neighbor-mean imputation."*

Response. We added a learned GNN imputer: a masked autoencoder with four GraphSAGE-style layers on the measured-only graph, trained on the training split to reconstruct the voltages of randomly dropped nodes, and used with the same dropped nodes as the mean imputer. Its reconstruction error is approximately 18 to 20 times lower than that of the mean imputer (RMSE 0.012–0.017 p.u. against 0.24–0.30 p.u.), but its macro-F1 at $k = 10$ is only 0.187, and its gains over mean imputation and over zeroing are not significant after Holm correction. Dropout-aware training exceeds it by 0.486 macro-F1 at $k = 10$ ($p_{\text{Holm}} = 0.037$; new row of Table 7). We report openly that the imputer did not fully converge for two of three seeds, that it was used only with a standard-trained classifier, and that joint training of imputer and classifier was not examined. We did not reimplement the full pipeline of [10], because it assumes that the set of unmeasured buses is fixed and known when the imputer and the classifier are built, which an inference-time failure does not provide.

Change. Section 4.3, imputation paragraph ([p. X: s4.3-imputation]); Table 6, new row ([p. X: tab6]); Table 7, new row ([p. X: tab7]); Section 5.2 ([p. X: s5.2]); Section 5.3 ([p. X: s5.3-imputation]).

Comment 2.5

> *"The four-layer depth was chosen for RGSAGE-Max; discuss its potential effect on comparisons with other architectures."*

Response. We agree. We now state that the depth was selected on RGSAGE-Max alone and then applied to every architecture, so it can favor RGSAGE-Max in the clean-accuracy ranking. We also state that the cross-architecture rankings are conditional on this depth, while the main claims (the dropout-aware gain, its clean-accuracy cost, and the comparison with SDR-STGNN) are each made within one architecture, between models of identical depth. The depth choice is now also listed as a limitation.

Change. Section 3.2, "Spatial depth" ([p. X: s3.2-depth]); Section 5.1 ([p. X: s5.1-depth]); Section 5.3 ([p. X: s5.3-depth]).

Part C — Reviewer #3

We thank Reviewer #3 for the careful comments on the presentation and the statistics.

Comment 3.1

> *"The interpretation of zeroed measurements and availability masks is important and should be emphasized in the methods."*

Response. We added a named paragraph, "Encoding of missing measurements," to Section 3.1. It defines the mask, states that a dropped sensor has all six channels set to zero, explains why a zero voltage alone is

ambiguous, and states that the mask is static in the clean corpus (only 14 of the 25 μPMUs are three-phase). Section 3.4 now refers back to this paragraph.

Change. Section 3.1, new paragraph ([p. X: s3.1-encoding]); Section 3.4 ([p. X: s3.4-missing]).

Comment 3.2

> *"Please verify the statistical claims and consistently report adjusted p-values where multiple comparisons are made."*

Response. We recomputed every test from the per-seed results. Every p-value in the manuscript is now Holm-adjusted (p_{Holm}) within a stated family, and Section 4.5 states the family rule once: each family contains the tests that answer one question. This changed two conclusions, which we now report faithfully: for standard RGSAGE-Max, the larger damage of targeted (worst-case) loss at $k = 1$ and $k = 2$ is consistent across seeds but not significant after correction ($p_{\text{Holm}} = 0.052$ and 0.067), and the dropout-aware worst-case score exceeds the standard average-case score significantly only at $k = 5$ and $k = 10$. The main claims remain significant (for example, the dropout-aware gain at $k = 10$, $p_{\text{Holm}} = 0.004$). Table 7 now contains seven tests, including the new comparison with the learned imputer, and all its values were recomputed.

Change. Abstract ([p. X: abstract]); Section 4.3 ([p. X: s4.3-targeted]); Section 4.5 and Table 7 ([p. X: s4.5-p2], [p. X: tab7]); Section 5.1 ([p. X: s5.1-p1]); Section 5.2 ([p. X: s5.2]); Section 6 ([p. X: s6]).

Comment 3.3

> *"Explain more precisely how the 26-class macro-F1 is calculated from window-level predictions."*

Response. We added a paragraph, "Macro-F1 computation," to Section 3.4. Each test window receives one prediction by soft voting over its 25 nodes; the predictions of all 247,000 test windows are pooled without aggregation over events; and the per-class F1 scores of the 26 classes are averaged without weighting (with the formula). The no-fault class holds 50% of the windows but only $1/26$ of the weight. We also state how the node-level macro-F1 is computed.

Change. Section 3.4, new paragraph ([p. X: s3.4-macro]).

Comment 3.4

> *"The simulation uses RMS phasor data; state this limitation prominently when discussing real-time fault localization."*

Response. We now state the limitation in the Abstract, the Discussion, the Limitations and the Conclusion: the data are quasi-static RMS phasors without electromagnetic transients, traveling waves, DC offset or sub-cycle content, and real-time localization on field μPMU streams is not demonstrated.

Change. Abstract ([p. X: abstract]); Section 5.1 ([p. X: s5.1-practical]); Section 5.3 ([p. X: s5.3-rms]); Section 6 ([p. X: s6]).

Comment 3.5

> *"Some claims about operating with 40% sensor loss should be qualified because results come from one simulated feeder."*

Response. We qualified every such claim: the result holds on the simulated IEEE 123-bus feeder under random, independent loss. We also changed "supports resilient monitoring" to "suggests that the augmentation can support resilient monitoring," and added that field validation is needed.

Change. Abstract ([p. X: abstract]); Section 5.1 ([p. X: s5.1-practical]); Section 6 ([p. X: s6]).

Part D — Editor

> *"Before the submission of the revised manuscript, the English language of your manuscript must be proofread and edited. Strictly follow Procedia Computer Science template."*

Response. English. The authors proofread and edited the English of the full revised manuscript. **Template.** The manuscript uses the Procedia Computer Science LaTeX template (elsarticle with ecrc); the header, footer, margins, captions and equation numbering follow the template. **Similarity.** The similarity index of the revised manuscript is [X]%; the report is attached as a supplementary file. This letter is included in the same file as the revised manuscript.

Part E — Additional change made by the authors

While revising, we found that the description of the data partition could be read as stronger than it is. The 70/15/15 partition is taken over events, not over fault scenarios: each of the 825 fault scenarios contributes events to the training and test splits, so the test events are new load realizations of the same fault scenarios. We now state this in Section 3.1 and list it as a limitation in Section 5.3 (generalization to unseen fault scenarios is not measured).

Change. Section 3.1 ([p. X: s3.1-partition]); Section 5.3 ([p. X: s5.3-scenario]).

Part F — Text removed or shortened (not visible in blue)

To keep the length close to the original despite the added content, we removed or shortened text that repeated other parts of the manuscript. No result and no answer to a reviewer comment was removed.

Location	Change	Where the content remains
Section 3.5, Algorithm 1	Removed	Its specific details (masking random generator reseeded every epoch, nodes drawn per window) are now one sentence in Section 3.5 ([p. X: s3.5]); the per-batch dropout level was already in the text.
Table 3	"Model configuration" and "Evaluation and reporting" rows removed	Sections 3.2 and 3.4; the one value only in Table 3 (RGATv2 attention dropout 0.3) is now in Section 3.2.
Section 3.7	"Objective" bullet removed	Section 3.2 (balanced corpus, no loss weighting).
Section 3.2, paragraph 1	Sentence listing the training method and SDR-STGNN removed	Introduction, paragraph 4.
Section 3.1, paragraph 4	Opening sentence ("Two encoding choices . . . ") removed; the paragraph was split	Section 3.1, two paragraphs.
Section 3.4, missingness model	"This reproduces the dataset's encoding of an absent sensor" replaced by a reference	Section 3.1.
Section 4.3, persistent dropout	Paragraph shortened; individual AUDC and retention values removed	The null result, the test result and the explanation remain.
Section 4.3, imputation and training grid	Numbers moved from the text to Table 6	Table 6.
Section 4.5, bootstrap	Shortened to one sentence	Section 4.5.
Section 5.1, paragraph 3	Repeated numbers removed	Sections 4.2 and 4.3.
Section 5.2, paragraph 1	Description of the mask channels shortened	Section 3.1, "Encoding of missing measurements."
Section 2	Paragraph 1 shortened; last sentence of the final paragraph removed	Introduction, paragraph 4.
Introduction	Paragraphs 1, 2 and 4 shortened	Sections 2, 3.2 and 3.7, and Table 2.
Fig. 1	Width reduced from 0.70 to 0.60 of the text width	—

We hope that the revised manuscript is now suitable for publication, and we thank the Editor and the Reviewers again for their time.

Sincerely,
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